

IP-SAKTI SAHAYAK

Family 11 — Brahmi / Bacopa cluster

Anchor: *Bacopa monnieri* (Brahmi)

Objective: Build a TKDL-only family cluster around Brahmi and use botanical proximity plus TKDL evidence to surface potential alternative candidates for further research.

Strict TKDL rule: The anchor and all candidate plants below are explicitly present in the official TKDL plant-name index. Because the public TKDL index contains only one *Bacopa* species, the related candidates are selected from other plants in the same modern botanical family (Plantaginaceae) that are also explicitly listed by TKDL. They are **not** presented as interchangeable with *Bacopa monnieri*.

1. Anchor / main plant

Bacopa monnieri — TKDL Ayurvedic name: Br■hm■; Unani/common entry: Brahmi; regional names include Neerpirami, Sapthalai, Sarumam and Sambiranipoond. TKDL also gives “coastal waterhyssop” among its common-name fields.

Main-component note: The public TKDL plant index does not provide a standardized active-constituent field for *Bacopa monnieri*. Therefore this PDF does not inject an external phytochemical claim into the TKDL dataset.

2. Related TKDL candidates — 4 verified species + 1 TKDL species-group entry

TKDL plant	Relationship to Bacopa	Similarity*	Evidence status
<i>Plantago asiatica</i> Linn.	Same modern family: Plantaginaceae; different genus	LOW-MED — 0.55	TKDL plant-index verified; use/formulation match required
<i>Plantago lanceolata</i> Linn.	Same modern family; different genus	LOW-MED — 0.55	TKDL plant-index verified; use/formulation match required
<i>Plantago major</i> Linn.	Same modern family; different genus	LOW-MED — 0.57	TKDL plant-index verified; use/formulation match required
<i>Plantago ovata</i> Forsk.	Same modern family; different genus	MEDIUM — 0.60	TKDL plant-index verified; formulation/use evidence to verify
<i>Plantago</i> spp.	TKDL species-group entry; not a single species	PROVISIONAL — 0.45	Index node only; not a substitution candidate

*Prototype screening score only; it is not a scientific equivalence score or legal clearance.

3. Official TKDL verification

Official TKDL Plant Name Index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The official TKDL index explicitly lists *Bacopa monnieri* as Brahmi and separately lists *Plantago asiatica*, *Plantago lanceolata*, *Plantago major* and *Plantago ovata*. It also contains a *Plantago* spp. entry. ■cite■turn0search0■turn4search0■

4. Taxonomy note

The TKDL public plant index is a nomenclature/search resource and does not itself provide a modern botanical-family label for every entry. For this prototype, Plantaginaceae is the modern taxonomy layer applied to the TKDL-listed species. Preserve the exact TKDL botanical names and synonyms in the source fields.

5. TKDL evidence — Brahmi

1118/CHE/2010: TKDL pre-grant opposition brought prior-art references on *Bacopa monnieri* (Brahmi), *Withania somnifera*, *Convolvulus pluricaulis*, *Asparagus racemosus* and *Tribulus terrestris* for nervous-system conditions, general weakness, loss of intellect and memory loss, and for intellect-promoting, memory-enhancing, anti-ageing and antioxidant uses. The report specifically cites a *Bacopa monnieri* single-ingredient formulation for memory enhancement/general tonic and another for memory enhancement and immunopotentiating/anti-ageing/antioxidant use. The examiner found the claimed combination obvious in view of the cited TKDL teaching; examination remained under process.

Official TKDL record: https://www.tkdil.res.in/tkdil/LangFrench/common/Examinarreport/reports/1118_CHE_2010.asp

1167/CHE/2009: TKDL examination material identifies *Bacopa monnieri* and *Centella asiatica* among herbs used for intellect-promoting, brain-tonic, memory-enhancing and nervine purposes. It cites a *Bacopa monnieri* + *Acorus calamus* formulation (“Medha vartak yog”) for intellect promotion and a *Bacopa monnieri* formulation for brain/memory-related disorders. The examiner found the claims lacking inventive step and falling under Section 3(p).

Official TKDL record: https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/1167_CHE_2009.asp

2271/CHE/2010: The official TKDL examination PDF states that *Bacopa monnieri* was disclosed for neurodegenerative disorders including Alzheimer’s disease, and that compositions containing *Bacopa monnieri* were cited for senile dementia and schizophrenia. The report concluded that the relevant claims lacked novelty and inventive step.

Official TKDL PDF: <https://www.tkdil.res.in/tkdil/Outcomes/2271CHE2010.pdf>

EP2266586: TKDL third-party observations cited *Bacopa monnieri* and *Centella asiatica* for longevity, cardio-tonic, memory-enhancer, immunopotentiating, anti-ageing/antioxidant and neurological/psychosis-related uses. The application was subsequently deemed withdrawn and closed.

Official TKDL record: <https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/EP2266586.asp>

6. Related-candidate interpretation

Plantago ovata: A verified same-family TKDL node and a useful first candidate for formulation/use similarity retrieval. It must not be called a Bacopa substitute solely because of family membership.

Plantago major, P. lanceolata and P. asiatica: All are separately listed in the official TKDL index. Treat them as lower-confidence family-neighbor nodes until exact TKDL formulation/use evidence relevant to the user's indication is attached.

Plantago spp.: Keep as an index/taxonomy node rather than a user-facing substitute because it is not a single species.

7. USP / UVP implementation

Family-Aware Alternative Discovery: Risk detected → botanical family mapped → TKDL-listed family neighbours retrieved → formulation/use similarity scored → ranked research candidates surfaced → separate patent/legal search triggered.

Example: For a Bacopa monnieri memory-enhancement formulation, IP-SAKTI can show Plantago ovata, P. major, P. lanceolata and P. asiatica as provisional same-family candidates, while clearly explaining that botanical family similarity does not establish equivalent efficacy or patent freedom.

8. Model-ready fields

family | anchor_plant | candidate_plant | botanical_name | TKDL_name/synonyms | plant_part | traditional_use | formulation | formulation_role | source_text | TKDL_case_id | official_TKDL_url | similarity_score | similarity_reasons | patent_IP_flag | evidence_level | verification_status

9. AI / legal guardrails

- Same family ≠ interchangeable.
- Same family ≠ freedom from patent infringement.
- Never output “no legal barriers” without claim-level patent analysis.
- A traditional-use record does not automatically prove a patent claim is invalid.
- Keep botanical identity and TKDL terminology exact.
- Separate TKDL evidence from novelty, inventive step and Section 3(p).
- Index-only candidates must be labelled as low/provisional evidence.
- Recommended wording: “**Potential alternative candidate for further research — TKDL-listed and family-related; botanical, formulation, efficacy and patent/legal verification required.**”